

Lecture 15

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MA 5104: Statistical Inference and Learning
Module 2: Estimation
Lecture 15: Resampling Methods for Statistical Inference

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What are we trying to approximate?

Suppose $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} F$ and we are interested in a parameter θ .
From the observed data, we calculate an estimator

$$\hat{\theta} = T(X_1, \dots, X_n).$$

Examples include

$$\bar{X}, \quad \text{median}(X_1, \dots, X_n), \quad \widehat{\text{cor}}(X, Y).$$

The estimate $\hat{\theta}$ would change if we collected a different random sample.
Therefore, to quantify uncertainty, we would like to understand

the sampling distribution of $\hat{\theta}$.

How can we approximate this distribution when its analytic form is difficult to derive?

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When the sampling distribution is known

Sometimes the sampling distribution can be derived exactly.

For example, if $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$, then

$$\bar{X} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right).$$

Therefore,

$$\text{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

We can use this distribution to construct confidence intervals.

But consider statistics such as $\text{median}(X_1, \dots, X_n)$, or $\text{mode}(X_1, \dots, X_n)$,

In general, the sampling distribution may be difficult to derive analytically.

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The basic resampling idea

We observe $x_1, \dots, x_n$. The population distribution F is unknown.
But the observed data provide an approximation to F .

The simplest approximation gives each observed value probability $\frac{1}{n}$.

This defines the **empirical distribution**:

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n I(x_i \leq x)$$

Thus,

$$P_{\hat{F}_n}(X = x_i) = \frac{1}{n}.$$

The basic idea behind resampling is therefore going from unknown population F $\rightarrow$ empirical population $\hat{F}_n$.

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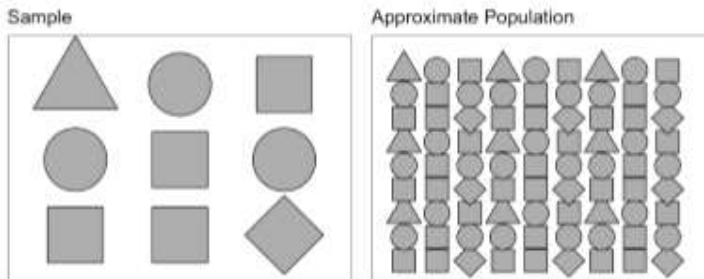
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Idea for the population

Suppose we were to sample from a population of shapes, and we observed 4/9 of the sample were squares, 3/9 were circles, and a triangle and a diamond. Then our best guess of what the population that we sampled from was a population with 4/9 squares, 3/9 circles, and 1/9 of triangles and diamonds.



The plug-in principle

Suppose the parameter can be written as a functional of the population distribution $\theta = t(F)$. The empirical distribution gives the natural estimator $\hat{\theta} = t(\hat{F}_n)$.

This is the **plug-in principle**:

$$F \longrightarrow \hat{F}_n \quad \text{and therefore} \quad t(F) \longrightarrow t(\hat{F}_n).$$

The bootstrap takes this idea one step further.

Instead of using $\hat{F}_n$ only to estimate θ , we use $\hat{F}_n$ to approximate the distribution of the estimator itself.

$$\mathcal{L}_F(\hat{\theta}) \approx \mathcal{L}_{\hat{F}_n}(\hat{\theta}^*)$$

The bootstrap principle

A bootstrap sample is generated by $X_1^*, \dots, X_n^* \stackrel{\text{iid}}{\sim} \hat{F}_n$. *(empirical dist)*
Equivalently, we sample n observations with replacement from

$$\{x_1, \dots, x_n\}.$$

For each bootstrap sample, calculate

$$\hat{\theta}^* = T(X_1^*, \dots, X_n^*).$$

Repeat this B times:

$$\hat{\theta}^{*1}, \hat{\theta}^{*2}, \dots, \hat{\theta}^{*B}.$$

These values form the **bootstrap distribution**.

$$\text{Sampling distribution} \approx \text{Bootstrap distribution}$$

Why sample with replacement?

Since $P_{\hat{F}_n}(X = x_i) = \frac{1}{n}$, each bootstrap observation is an independent draw from the empirical distribution.

Thus an observation may appear zero times, once, or multiple times.

For example, if $x = \{2, 4, 5, 7, 9\}$, then $x^* = \{4, 4, 2, 9, 7\}$ is a valid bootstrap sample.

If N_i denotes the number of times x_i appears in one bootstrap sample, then

$$N_i \sim \text{Binomial}\left(n, \frac{1}{n}\right).$$

For large n , $N_i \approx \text{Poisson}(1)$.

Hence the bootstrap deliberately creates samples that differ from the original sample while retaining its empirical distribution.

Bootstrap distribution: what does it tell us?

After B bootstrap replications, we have $\hat{\theta}^{*1}, \dots, \hat{\theta}^{*B}$.

We can study the center, spread, and shape of this distribution.

The bootstrap estimate of standard error is

$$\bar{\theta}^* = \frac{1}{B} \sum_{b=1}^B \hat{\theta}^{*b}, \text{ and}$$

$$\widehat{SE}_{\text{boot}} = \sqrt{\frac{1}{B-1} \sum_{b=1}^B (\hat{\theta}^{*b} - \bar{\theta}^*)^2}.$$

The bootstrap distribution can also reveal skewness, heavy tails, multimodality, boundedness, etc.

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Bootstrap and bias

Recall the definition of bias: $\text{Bias}(\hat{\theta}) = \mathbb{E}_F(\hat{\theta}) - \theta$.

The bootstrap replaces the unknown population distribution F by $\hat{F}_n$.
Thus,

$$\mathbb{E}_F(\hat{\theta}) \rightarrow \mathbb{E}_{\hat{F}_n}(\hat{\theta}^*).$$

The bootstrap estimate of bias is therefore

$$\widehat{\text{Bias}}_{\text{boot}} = \bar{\theta}^* - \hat{\theta}.$$

A bias-adjusted estimate would be

$$\hat{\theta}_{\text{adj}} = \hat{\theta} - \widehat{\text{Bias}}_{\text{boot}} = 2\hat{\theta} - \bar{\theta}^*.$$

However, estimating bias does not imply that we should correct for it. The estimated bias itself can be noisy.

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Bootstrap does not fix a bad estimator

The bootstrap repeatedly applies the same estimation procedure to resampled data.

Therefore, if the original procedure is systematically biased because of model misspecification, the bootstrap does not automatically remove that problem.

For example, suppose the true relationship is nonlinear: $Y = f(X) + \varepsilon$ but we fit a linear regression model. Bootstrapping the fitted procedure gives repeated estimates of the **linear approximation**.

Thus, bootstrap variability $\neq$ validity of the estimator.

The bootstrap can answer: "How variable is my estimator?"
without answering: "Is my estimator targeting the right quantity?"

Jackknife: leave one observation out

The jackknife provides another way of perturbing the observed data.
Start with $\hat{\theta} = T(x_1, \dots, x_n)$.

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Start with $\hat{\theta} = T(x_1, \dots, x_n)$.

For each i , remove observation x_i and calculate

$$\hat{\theta}_{(-i)} = T(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n).$$

This produces $\hat{\theta}_{(-1)}, \hat{\theta}_{(-2)}, \dots, \hat{\theta}_{(-n)}$.

Their average is

$$\bar{\theta}_{(-)} = \frac{1}{n} \sum_{i=1}^n \hat{\theta}_{(-i)}.$$

The jackknife asks:

How sensitive is $\hat{\theta}$ to individual observations?

Jackknife standard error

The jackknife estimate of standard error is

$$\widehat{SE}_{\text{jack}} = \sqrt{\frac{n-1}{n} \sum_{i=1}^n (\hat{\theta}_{(-i)} - \bar{\theta}_{(-)})^2}.$$

The jackknife estimate of bias is

$$\widehat{\text{Bias}}_{\text{jack}} = (n-1) (\bar{\theta}_{(-)} - \hat{\theta}).$$

The jackknife is particularly useful for understanding how individual observations influence an estimator.

Jackknife = systematic leave-one-out perturbation.

Jackknife versus bootstrap

	Jackknife	Bootstrap
Basic operation	Delete one observation	Sample with replacement
Number of replicates	n	B , usually much larger than n
Nature	Deterministic	Monte Carlo
Main idea	Local perturbation	Repeated resampling
Standard error	Yes	Yes
Bias	Yes	Yes
Approximate full distribution	Limited	Yes ✓

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When can the ordinary bootstrap be problematic?

The bootstrap is not automatically valid for every statistic. It works particularly well when the statistic is sufficiently smooth as a function of the underlying distribution.

Problems can arise for statistics such as $\max(X_1, \dots, X_n)$, or median, where the sampling distribution can be determined primarily by extreme observations.

Similarly, for some nonsmooth statistics and small samples, the bootstrap distribution can be a poor approximation to the true sampling distribution.

Therefore,

Bootstrap validity depends on both the data and the statistic.

A bootstrap histogram should not automatically be interpreted as proof that the bootstrap approximation is accurate.

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From bootstrap distributions to confidence intervals

Suppose we want a confidence interval for θ .

The bootstrap gives us an approximation to the sampling distribution of $\hat{\theta}$.

The basic problem is to use this approximate distribution to infer where the unknown θ could lie.

The key distinction is between $\hat{\theta} - \theta$ and the bootstrap analogue $\hat{\theta}^* - \hat{\theta}$.

The bootstrap uses the latter to learn about the former.

This leads to several different bootstrap confidence intervals.

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Percentile $\rightarrow$ Basic $\rightarrow$ Bias-corrected $\rightarrow$ Bootstrap-t

The percentile bootstrap interval

Let $q_{\alpha/2}^*$ and $q_{1-\alpha/2}^*$ be the corresponding empirical quantiles of the bootstrap estimates $\hat{\theta}^{*1}, \dots, \hat{\theta}^{*B}$.

The percentile confidence interval is

$$\left[q_{\alpha/2}^*, q_{1-\alpha/2}^* \right].$$

For a 95% interval, $[q_{0.025}^*, q_{0.975}^*]$.

An attractive feature is that the interval automatically reflects the shape of the bootstrap distribution.

If the bootstrap distribution is skewed, the interval need not be symmetric around $\hat{\theta}$.

The basic bootstrap interval

The percentile interval uses the bootstrap estimates directly.

Another approach starts from $\hat{\theta} - \theta$. The bootstrap analogue is $\hat{\theta}^* - \hat{\theta}$.

Suppose the **bootstrap error** has quantiles $q_{e,\alpha/2}^*$ and $q_{e,1-\alpha/2}^*$.

Then reversing the bootstrap error gives

$$\left[\hat{\theta} - q_{e,1-\alpha/2}^*, \hat{\theta} - q_{e,\alpha/2}^* \right] = \left[2\hat{\theta} - q_{e,1-\alpha/2}^*, 2\hat{\theta} - q_{e,\alpha/2}^* \right].$$

This is the **basic** or **reverse-percentile** interval.

The distinction is:

Percentile: use bootstrap estimates

Basic: use bootstrap errors and reverse them

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Bootstrap- t intervals

Consider the standardized statistic $T = \frac{\hat{\theta} - \theta}{\text{SE}(\hat{\theta})}$.

In classical inference, we may know or approximate the distribution of T .

The bootstrap provides another possibility.

For each bootstrap sample calculate

$$T^* = \frac{\hat{\theta}^* - \hat{\theta}}{\widehat{\text{SE}}^*} \approx \hat{\theta} - \theta$$

Use the bootstrap distribution of T^* to obtain quantiles

$$t_{\alpha/2}^*, \quad t_{1-\alpha/2}^*.$$

The resulting confidence interval is

$$\left[\hat{\theta} - t_{1-\alpha/2}^* \widehat{\text{SE}}, \hat{\theta} - t_{\alpha/2}^* \widehat{\text{SE}} \right].$$

SE is from one original data
 $\approx \hat{\theta}$ data is same

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Why studentize?

The percentile method works with $\hat{\theta}^*$.

The bootstrap- t method works with $\frac{\hat{\theta}^* - \hat{\theta}}{SE^*}$.

The standardization is important because the variability of $\hat{\theta}$ may itself depend on the parameter.

Thus the bootstrap- t approach attempts to approximate the sampling distribution of a more stable quantity:

estimation error $\rightarrow$ standardized estimation error.

The trade-off is computational: every bootstrap sample requires its own standard-error estimate.

This is one reason the bootstrap- t method can be substantially more computationally demanding.

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Comparing bootstrap confidence intervals

Interval	Main idea
✓ Percentile	Use quantiles of the bootstrap estimates
Basic	Use quantiles of the bootstrap estimation error and reverse them
✓ Bootstrap- t	Estimate the distribution of a <u>studentized</u> statistic

There is no single bootstrap confidence interval that is universally best.

The appropriate method depends on the statistic, sample size, skewness, bias, and computational considerations.

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Three ways to quantify uncertainty

	Object	Source of uncertainty
Frequentist theory	Sampling distribution of $\hat{\theta}$	Repeated sampling from F
Bayesian inference	Posterior $\pi(\theta x)$	Uncertainty about θ after observing x
Bootstrap	Approximate sampling distribution of $\hat{\theta}$	Resampling from $\hat{F}_n$

≠ frequentist CI

Note that the Bayesian credible interval $\neq$ bootstrap confidence interval.

They may look similar numerically, but they are based on different probability statements.

Take-home messages

- The fundamental object for frequentist uncertainty is the sampling distribution of an estimator.
- The jackknife systematically perturbs the data by deleting one observation at a time.
- The bootstrap treats the empirical distribution as an approximation to the population and repeatedly resamples from it.
- The bootstrap can estimate standard errors, bias, and the shape of sampling distributions.
- The bootstrap quantifies uncertainty in an estimator; it does not automatically make a poor estimator or model valid.